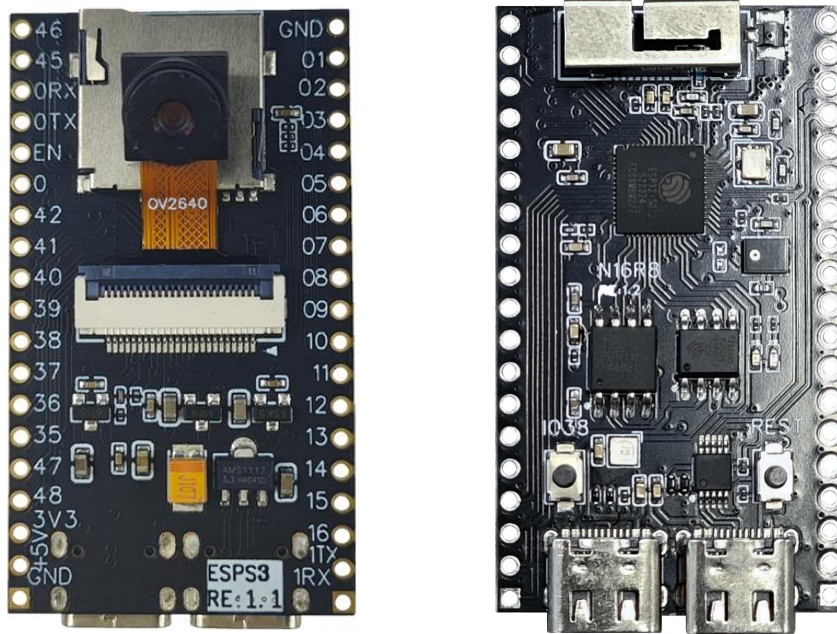


ESP32-S3 MINI

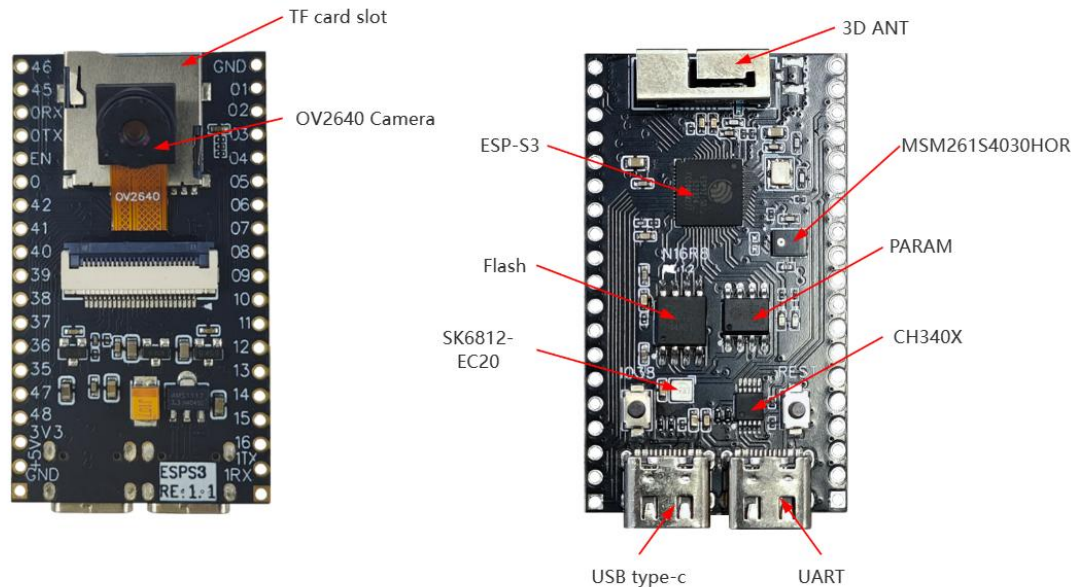
products specification



Features:

The ESP32-S3 Mini is a feature-rich, compact and portable Internet of Things (IoT) development board. This development board is equipped with a Tensilica Xtensa LX7 core processor with a main frequency of up to 240 MHz and a MSM261D3526H1CPM microphone. It is furnished with 512 KB of on-chip RAM (SRAM) for storing program codes and data, as well as external flash and external PSRAM expansion to meet the storage requirements of larger capacity. It supports Wi-Fi and Bluetooth communication and offers an abundance of peripheral interfaces, including multiple Universal Serial Buses (SPI, I2C, UART), GPIO pins, ADC, and DAC. These interfaces can be utilized for connecting various sensors, actuators, and external devices. This development board supports programming using the Arduino development environment and also provides the ESP-IDF development framework, enabling more low-level programming and system debugging. It is suitable for prototyping and the rapid development of various IoT applications, such as smart homes, sensor networks, and remote monitoring, etc.

Module hardware layout



Functions Overview

MCU : ESP32-S3	Xtensa®32LX7 dual-core processor, frequency adjustable between 2040MHz, integrated 2.4GHz Wi-Fi and Bluetooth dual-mode, 40nm process, integrated 512KB RAM and 384KB ROM memory (for program startup and core function call) FLASH support SPI, DualSPIQuadSPI, OctalSPI and RAM peripherals.
USB : CH340X	Automatic download circuit, with E8051 core compatible with MCS51 instruction set, the average instruction speed is 8~15 times faster than the standard MCS51, ESP32-CAM module uses CH340X to realize automatic download circuit, with USB can be easily written and debugged ESP32.
MSM261D3526H1CPM	It is equipped with a 24-bit I2S interface, featuring PDM digital output mode and a flexible signal processing architecture. It supports independent use (only analog output) or deep integration with digital systems (via I2S/PDM), adapting to diverse application scenarios such as smart speakers and IoT devices.

ESP32-S3 MINI specification

TF card slot.	The module has a TF card slot on board, which is compatible with most standard TF cards on the market, supports up to 16GB, and can realize functions such as photo storage, and playback with the camera.
SK6812-EC20	The SK6812-EC20 is a digitally programmable color LED with integrated LED and control circuitry. It is available in a surface mount technology (SMD) package for compact size and reliable performance. The SK6812-EC20 LED is a digitally programmable RGB LED.
OV2640 Camera	Provides a complete power over the serial camera control bus (SCCB) interface for single-chip JXGA (1632x1232) cameras and image processors, providing YUV(422/420)/YCbCr422 full-frame sampling, scaling, or windowing of 8-bit/10-bit images. THE BEST IMAGE DISTANCE IS 20-250CM.
TYPE-C 16PIN 2MD	Two electrical connectors for Type-C interface devices are available, with high-speed transmission, reversible pluggability, and multifunctional expansion. Among them, USB2 is automatically downloaded with CH340 connection. USB1 is connected to the ESP32-S3 chip.

Operating Principle

Output Model Category:Digital sensor	How it works:Biosensors
Sensor Category:Optical sensor	Working current:0.15A Working voltage:5V

Support Software Development

C++ / Esp-IDF / Ardiuno-IDE / Vscope / Microphython

Module application scenarios

DIY secondary development

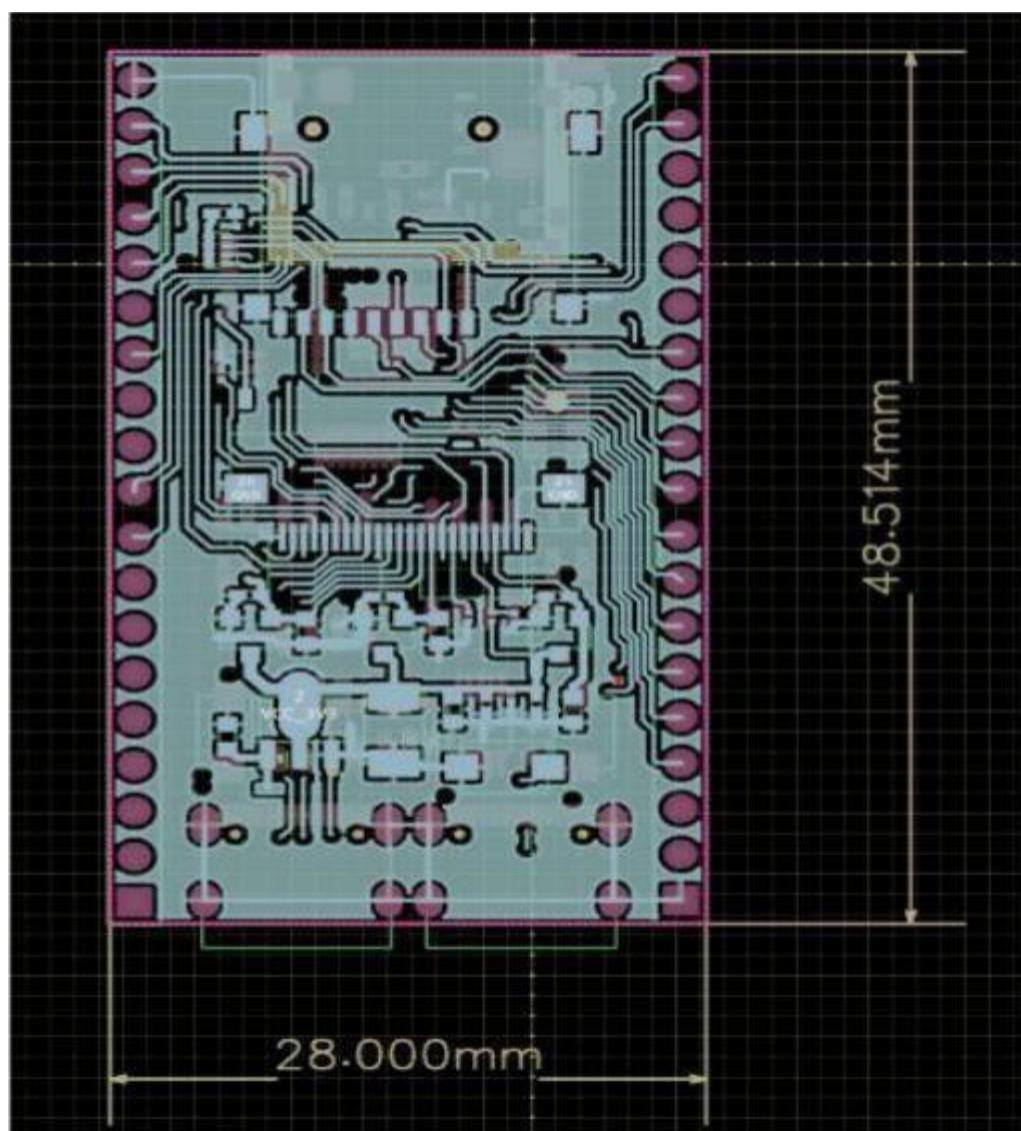
ESP32-S3 MINI specification

Low cost camera solution
Iot node devices

Size	
Length	48.51400mm
width	28.00mm

Document Update Record		
Version	Time	Mark
V1.0	2023-08-11	first release
V1.1	2023-11-30	

Module size diagram



Unit: mm